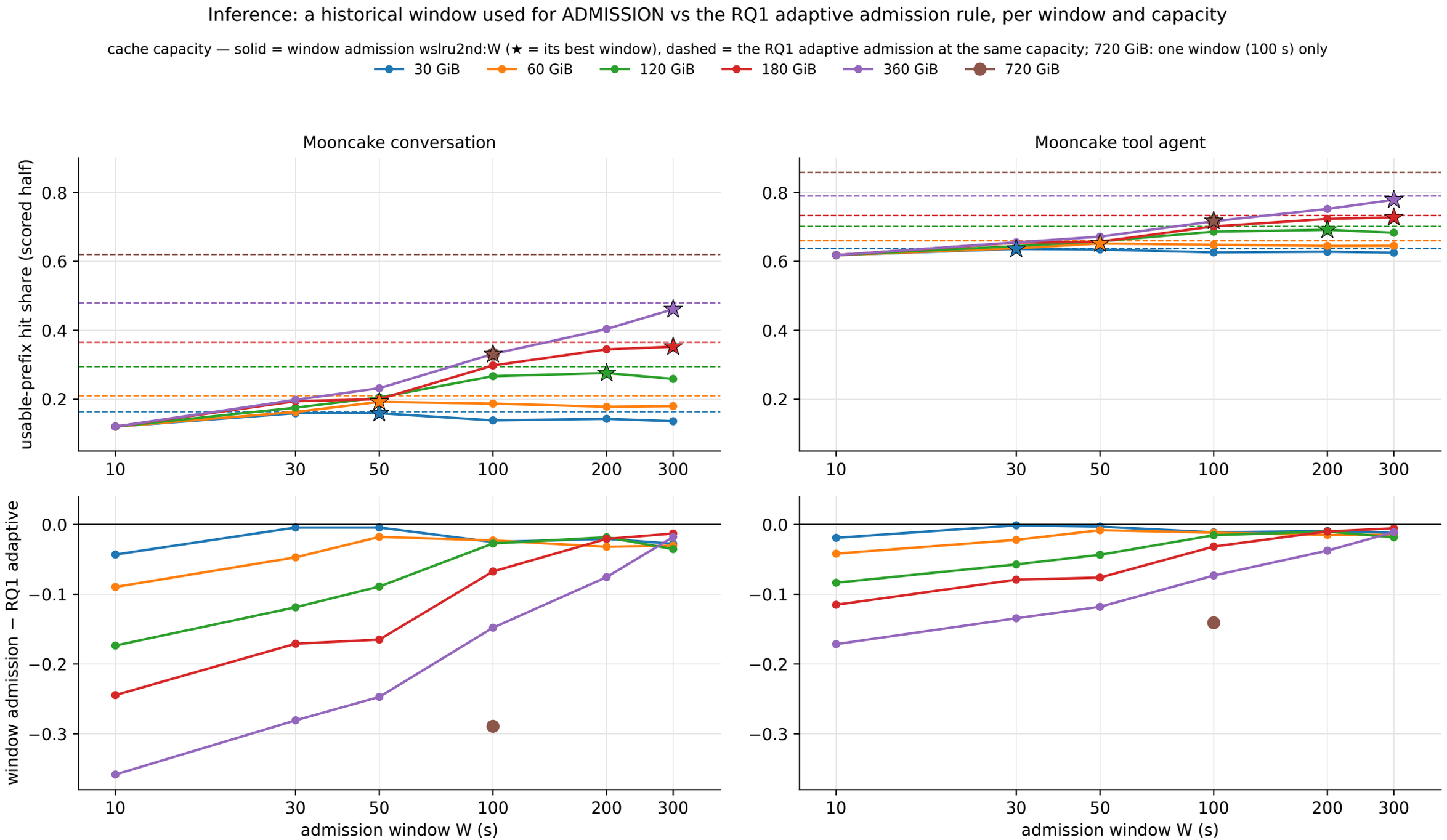

KV-cache admission replay (kv_admission_sim.py) of the Mooncake conversation and tool-agent traces; one host cache of 8B blocks (67.1 MB); the first half of each trace warms the cache, the second half is scored. Usable-prefix hit share = share of the scored block lookups with a reuse that hit AND whose whole preceding prefix in the request also hit (a hit the engine can use). wslru2nd:W = admit a block only if it was looked up within the last W s (the first block of a request is always admitted), with the wslru:W protection; RQ1 adaptive = 2nd_int_auto: admit on a second touch whose re-lookup gap ≤ the online residence estimate, SLRU structure, automatic switch to first-touch admission when the admit-all residence ≥ the median first-touch re-lookup gap.